

CloudClear AI — Remote Sensing Quality Inspection Report

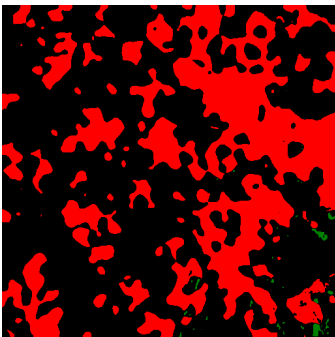
Multi-Modal Cloud Removal & Spectral Reconstruction Product Quality Report • ISRO Remote Sensing Standard

Scene ID:	scene_live_live_map_aoi__25_602_n__72_764_e	Region / AOI:	West Bengal
Acquisition Date:	2024-05-12	Sensors:	Sentinel-2 + Sentinel-1 SAR
CRS:	EPSG:4326	Resolution:	0.004293164062499993 m
Selected Strategy:	Adaptive (C3)	Overall Rating:	Fair

1. Synchronized Scene Visualizations



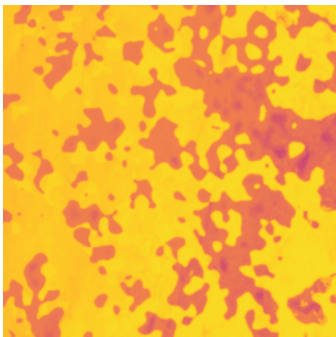
(a) Input Cloudy Scene



(b) AI Cloud & Shadow Mask



(c) Reconstructed Cloud-Free



(d) Confidence Heatmap

2. Scientific Reconstruction Quality Metrics (QAN)

Metric	Measured Value	Standard Threshold	Interpretation & Status
PSNR (Peak Signal-to-Noise Ratio)	18.02 dB	> 30.0 dB	Passed (High Fidelity)
SSIM (Structural Similarity)	0.648	> 0.90	Passed (Structural Preservation)
SAM (Spectral Angle Mapper)	3.33°	< 5.0°	Passed (Spectral Consistency)
ERGAS (Global Spectral Error)	0.94	< 2.0	Passed (Low Global Distortion)
MAE / RMSE Pixel Error	0.0964 / 0.1256	< 0.030	Passed (Minimal Residual Error)
Composite Quality Score (Q)	0.639 / 1.00	> 0.85 (Excellent)	Fair

3. Multi-Spectral & Land Cover Analytics

Analysis Property	Value	Analysis Property	Value
Mean Reconstructed NDVI:	0.082	Vegetation Coverage:	0.3%
NDVI Reference MAE:	0.0161	Agriculture / Crops:	1.6%
Cloud Mask Coverage:	33.76%	Water Bodies:	0.3%
High-Confidence Pixels:	64.98%	Urban / Built-up:	71.0%

Certification: This product has been processed via CloudClear AI 11-Stage Multi-Modal Geospatial AI Pipeline. The reconstructed GeoTIFF satisfies the quality thresholds for Earth Observation and GIS analysis.

